

CENTRIFUGAL PUMP P-102 CONDITION INSPECTION REPORT (JULY 2026)

1.0 Equipment Identification

Asset ID: P-102

Equipment Description: Centrifugal Crude Charge Pump

Unit: Unit 101 — Crude Distillation

Inspection Date: 2026-07-14

Lead Vibration Specialist: A. Verma

2.0 Measured Operating Parameters

Pump P-102 urgent assessment following vibration telemetry alert:

- Overall Vibration Velocity (RMS): 3.7 mm/s (ABNORMAL — Threshold Exceeded)
- Drive End Bearing Temperature: 77°C (Elevated)
- Non-Drive End Bearing Temperature: 74°C
- Discharge Pressure: 13.8 bar
- Acoustic Analysis: Audible bearing rumble detected

3.0 Findings & Recommendations

Pump P-102 — Vibration: 3.7 mm/s. Temperature: 77°C. Status: Abnormal. Measured overall vibration of 3.7 mm/s breaches the manufacturer attention threshold of 3.0 mm/s specified in Section 4.2 of the operating manual.

Increasing vibration: Jan 2.1 -> Apr 2.8 -> Jul 3.7 mm/s (+76%). Engineering inspection recommended. The evidence does NOT establish imminent catastrophic failure.